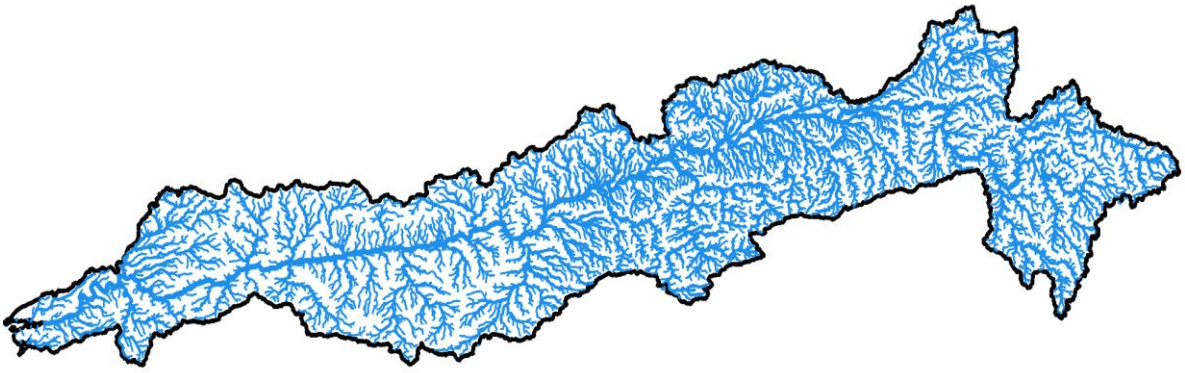
National River Conservation Directorate
Ministry of Jal Shakti, Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Management Plan for Industrial Wastewater



June 2026

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National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and the Ministry of Jal Shakti, provides financial assistance to the State Government for the conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies.

www.nrcd.nic.in

Centres for Narmada River Basin Management and Studies (cNarmada)

The Centre for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the centre serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement.

www.cnarmada.org

Centres for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as a think-tank in the implementation and dynamic evolution of the Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India.

www.cganga.org

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PREFACE

This report presents the comprehensive Management Plan for Industrial Wastewater in the Narmada River Basin. The Narmada River Basin supports millions of people across central and western India and is an important ecological and economic region.

Due to increased urbanization and industrialization in the basin, managing industrial effluents and emerging pollutants has become increasingly important for protecting the ecosystem.
This report assesses the status of industrial wastewater generation, treatment infrastructure, hazardous waste generation, and its management.

This report identifies the key challenges and gaps and provides recommendations for sustainable industrial practices. The report is intended to serve as a reference for policymakers, regulatory authorities, industries, stakeholders, and researchers working towards the conservation and sustainable management of the Narmada River Basin.

Centre for Narmada River Basin
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Abbreviations and Acronyms

Abbreviation	Full Form
AMRUT	Atal Mission for Rejuvenation and Urban Transformation
CAAQMS	Continuous Ambient Air Quality Monitoring System
CETP	Common Effluent Treatment Plant
CEQMS	Continuous Effluent Quality Monitoring System
CEMS	Continuous Emission Monitoring System
CPCB	Central Pollution Control Board
cGanga	Centre for Ganga River Basin Management and Studies
cNarmada	Centre for Narmada River Basin Management and Studies
DEPs	District Environmental Plans
FSTP	Faecal Sludge Treatment Plant
IPC	Industrial Pollution Control
JJM	Jal Jeevan Mission
MEE	Multi-Effect Evaporator
MLD	Million Litres per Day
MoHUA	Ministry of Housing and Urban Affairs
MoJS	Ministry of Jal Shakti
MPPCB	Madhya Pradesh Pollution Control Board
PPP	Public-Private Partnership
RO	Reverse Osmosis
RTM	Real-Time Monitoring
SBM-U	Swachh Bharat Mission – Urban
SMI	Small & Medium Enterprises
SPCB	State Pollution Control Board
STP	Sewage Treatment Plant
TSDF	Treatment, Storage, and Disposal Facility
ULB	Urban Local Body
ZLD	Zero Liquid Discharge

1 Introduction & Objectives

The Narmada River is one of the most significant west-flowing river systems in India, traversing the states of Chhattisgarh, Madhya Pradesh, Maharashtra, and Gujarat before draining into the Arabian Sea. The basin spans a large area and supports a population of approximately 20,936,840 making it a critical socio-economic and ecological region. The population distribution within the basin is spatially heterogeneous, with 8,603,425 people in the upper basin, 9,126,886 in the middle basin, and 3,206,529 in the lower basin (Demography Report; cNarmada, 2025).

The basin also supports a diverse and expanding industrial landscape that plays a vital role in regional economic development. Industrial activities across the Upper and Middle Basin regions include agro- and food processing, manufacturing, chemical industries, mining and mineral processing, energy generation, and waste and water management. Because many of these industries use the river as their primary source of water and/or require access to this water for other aspects of their operation(s), they are highly dependent upon sustainable water resources, and therefore, there exists a strong relationship between industry growth and water resource sustainability. In recent years, the establishment of numerous micro-, small-, medium-, and large-scale industries in various districts, including Jabalpur, Indore, Dhar, Sagar, Katni, Balaghat, Raisen, and Khandwa, has greatly increased environmental pressure (Industrial Profile of the Narmada basin, 2025). As industrial activity has increased, emerging pollutants/contaminants have become major factors affecting water quality and ecosystem health in the basin. Some examples of emerging pollutants generated directly by industrialization include industrial effluent discharges from factories, hazardous waste produced by industrial processes, emissions from factory chimneys, and contamination from mining operations. To address emerging environmental concerns, government agencies such as the Central Pollution Control Board (CPCB) and the Madhya Pradesh Pollution Control Board (MPPCB) have established environmental regulations and implemented real-time monitoring programs.

This report provides an overview of current industrial wastewater generation, the installed wastewater treatment infrastructure, and the practice of managing industrial effluents within the Narmada River Basin. It will identify gaps in treatment facilities, such as CETPs, ZLD

systems, and hazardous waste management infrastructure, using both district-level data and field visit data. In addition to identifying these gaps, this report describes and documents regulatory bodies' current monitoring practices and recommends a systematic approach to improve industrial effluent management through better treatment infrastructure, enhanced monitoring systems, and collaboration among all stakeholders.

2 Cluster Analysis

The industrial areas along the Narmada River Basin display an extremely diverse and regionalized form of industrialization, influenced by a combination of physical geographic factors, including location, natural resources, and infrastructure connectivity. This is demonstrated in *Figure 1,* where industrial estates have developed throughout much of the Narmada River Basin, creating a continuous belt of industry (industrial corridor) stretching from the heavily industrialized western regions of Gujarat to the developing or emergent industrial areas in the eastern sections of Madhya Pradesh.

2.1 Industrial Clusters

The Upper Narmada Basin has a variety of industrial areas, but they are scattered; however, all are strategically located near urban centers. Examples of these estates include Industrial Area - Adhartaal, Industrial Area Richhai, Industrial Area Bagaspur, and the Semi-Urban Industrial Estate Narsinghpur, which illustrates the industrial areas surrounding Jabalpur, which is considered one of the primary growth centers within the region. Additionally, other clusters exist in the regions of Sagar, Seoni, Mandla, and Damoh that represent localized industrial growth resulting from agro-based industry, mining/processing minerals, engineering units, and developing small-scale manufacturers. However, the close location of many of these industrial areas to the Narmada River highlights the necessity of sustainable planning practices, effective pollution controls, and ongoing environmental monitoring to protect the basin's ecosystem while allowing for continued long-term growth of its industries. The lower Narmada Basin has larger-scale industrial areas, including Bharuch, Ankleshwar, Jhagadia, Rajpipla, Valia, and the Industrial Area Bharuch. These areas are primarily involved in large-scale chemical, petrochemical, pharmaceutical, and manufacturing activities. Their locations on key national transportation corridors and ports provide them with access to raw materials and markets needed to facilitate growth. Eastward, into the middle basin, are smaller-scale industrial clusters, including Alirajpur, Anjad, Badwani, Sendhwa, Khargone, Bhikangaon, and Khandwa. These clusters have experienced stable growth due to increased involvement of agro-

processing, textile units, and small-scale manufacturing and mining-based industries. The existence of developed industrial estates such as Industrial Area - Indore Road (Khandwa), Industrial Area - Badiwah, and Industrial Area - Dolria indicates increasing connections with the nearby urban centers of Indore and Dhar (MPIDC).

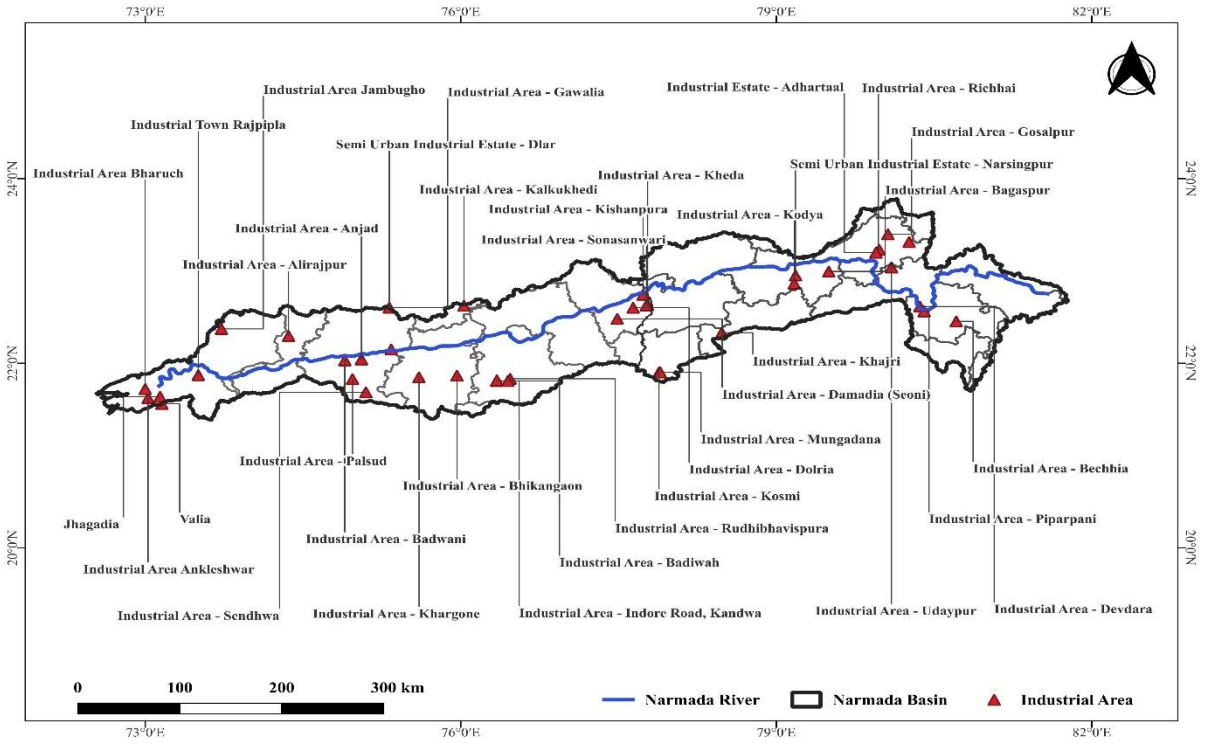


Figure 1: Spatial distribution of industrial areas across the Narmada River Basin

Source: Data Provided by MPIDC in 2025

2.2 Industrial Wastewater Generation and Treatment Status

The district-wise status for industrial wastewater generation, treatment, and compliance is shown in *Table 1and Table 2*. The data clearly indicate that industrial wastewater generation is unevenly distributed across the entire watershed, with substantial contributions from districts such as Jabalpur, Betul, Chhindwara, Khandwa, Khargone, and Sehore. Some districts in the region have adopted treatment methods such as ZLD or partial treatment; however, most districts in this study lack sufficient treatment infrastructure to treat their generated wastewater.

Jabalpur, located in the upper watershed, generates about 50 MLD of industrial wastewater, which is treated only partially at existing CETPs. Like Jabalpur, Betul, and Chhindwara, generate large amounts of industrial wastewater but lack sufficient local treatment infrastructure. Like these districts, Khandwa (39.60 MLD) and Khargone (22.28 MLD) in the middle watershed are also significant contributors to the total wastewater load. Many of these

districts have little or no treatment infrastructure available and/or significant non-compliance issues due to some industries discharging wastewater outside of standards.

Table 1: District-wise Industrial Wastewater Generation, Treatment Status, and Compliance across the Narmada Basin

*Source: District Environmental Plans 2025 for districts (*Data presented here is for the whole district)*

Basin	Districts	No. of industries discharging wastewater	Total Quantity of Industrial wastewater generated	Quantity of treated industrial wastewater discharged into Nalas / Rivers	Common Effluent Treatment Facilities	No. of industries meeting standards	No. of industries not meeting discharge standards
Upper	Anuppur	Nil	14.5 MLD	Nil	Nil	208	Nil
	Balaghat	30	16 MLD	Nil	Nil	30	Not Accessed
	Betul	Not assessed	52.18 MLD	Nil	None	96	Nil
	Chhindwara	NIL	31.51 MLD	Nil	Existing CETP's – 03 (Capacity – 8.7 MLD) in Bhopal, Indore and Burhanpur Newly Constructed CETP's – 02 (Capacity – 0.6 MLD) in Indore and Jhabua	-	-
	Damoh	2	0 MLD [ZLD Industries]	Nil	None	2	Nil

	Dindori	0 [No wastewater discharging industries in the district]	0 [No industrial wastewater generating industry]	None	None	114	None
	Narmadapuram*	Nil	Zero Liquid Discharge (ZLD)	Not Assessed	No CETFs are available with the district.	251	-
	Jabalpur	150	50 MLD	35 MLD	CETP – 2 (combined capacity of 25 MLD)	30	/
	Katni	Not Assessed	Not Assessed	0	0	Not Assessed	0
	Mandla	Nil	Nil	Nil	Nil	Nil	Nil
	Narsimhapur	15	10 MLD	None	None	15	None
	Raisen	Nil	13.71 MLD	Nil	Nil	699	-
	Sagar	None	None	None	None	78	None
	Seoni	28	10 MLD	Nil	Nil	28	Nil
	Umaria	0	0.17 MLD	None	None	128	0
Middle	Alirajpur	0	145.51 KLD	0	0	65	29
	Barwani	29	3 MLD	Not Assessed	Nil	23	29
	Bhopal	0 (All are ZLD)	6.21 MLD	Nil	1	36	None
	Burhanpur	61	4.5 MLD	None	01 CETP Sindhi Basti,	61	0

					Burhanpur (4.5 MLD)		
	Dewas	0	4.613 MLD	0	0	79 (No. of water polluting industries)	0
	Dhar	0 (All are ZLD)	33,515 KLD	Nil	0	1355	0
	Harda	Nil	878.7 KLD	Nil	Nil	80	None
	Indore	NA	NA	NA	NA	NA	NA
	Jhabua	All water-polluting industries having wastewater treatment facility.	20.2 MLD	Nil	1	193	Nil
	Khandwa	6	39.60 MLD	None	None	6	None
	Khargone	17	22.28 MLD	Not Assessed	There are no CETFs in the district.	92	79
	Sehore	Zero liquid Discharge	24.79 MLD	Nil	Nil	21	Nil

Table 2: Industrial Wastewater Generation in the Lower Narmada Basin (*Source: Gujrat Pollution Control Board, District Environmental Plans 2022).*

District	Industrial Waste water
Narmada	2.546 MLD
Chota udepur	0.10805MLD
Bharuch Region	126.787MLD
Ankleshwar Region	104.414MLD

The table 2 presents the quantity of industrial wastewater generated in different districts and industrial regions of the Lower Narmada Basin. It demonstrates the significant contribution of industrial activities, particularly in Bharuch and Ankleshwar, to the overall wastewater load.

1.1 Industrial Clusters and CETP Requirements

MPPCB divides industries into red, orange, and green. This depends on the level of pollution potential, and for controlling pollution, the MPPCB grants industries consent for environmental law enforcement. Based on the MPPCB industrial wastewater data for various districts of the Upper and Middle Narmada Basin, it is observed that there is no uniformity in wastewater management practices. Some districts, such as Dhar, Sehore, Narmadapuram, and Jhabua, have a zero-liquid-discharge system or an in-house facility for industrial wastewater management, ensuring minimal discharge into water bodies. Many districts generate large amounts of industrial wastewater, but only a few have CETP facilities.

Districts like Jabalpur have 2 CETPs with a capacity of 25 MLD each; Burhanpur has 1 CETP with a capacity of 4.5 MLD; and Bhopal, Indore, and Jhabua each have 1 CETP. In many districts, wastewater treatment for industrial discharges is either absent or inadequately assessed, despite significant wastewater generation. From the data, it is concluded that many industries rely on the industry-level treatment and ZLD practices and are limited to the CETP facilities. This gap highlights the need for a common treatment infrastructure, particularly in the industrial areas of the Narmada Basin's districts.

Districts like Balaghat, Chhindwara, Narmadapuram, and Sagar also have industrial areas, but no CETP facility is present. Similarly, districts like Barwani, Dewas, Khandwa, and Khargone, Sehore in the Middle Narmada basin, are without CETP facilities (MPPCB, 2025).

Existing industrial wastewater treatment facilities in the Narmada Basin

According to data for Jabalpur, approximately 150 industries generate industrial effluent totaling around 50 MLD, of which 35 MLD is treated effluent, ensuring it is free of harmful pollutants and meets environmental standards. In Jabalpur, there is a combination of Effluent and common effluent treatment plants that treat wastewater from different industries. This combined approach to the treatment plant is highly effective because a variety of industries are present in Jabalpur, including plywood, agro, paper, and plastic. The municipal corporation also focuses on adopting advanced technologies to improve wastewater treatment efficiency, thereby contributing to water sustainability in the districts and the Narmada Basin (MPPCB, DEP 2025).

In Jabalpur, the ETP plant is executed by the Amrita Water Solution Services.

The effluent treatment plant contains the following units:

Screening: for the removal of large particles

Neutralization: pH adjustment

Sedimentation: solids settling

Biological Treatment: breakdown of organic compounds by microorganisms.

In the ETP plant, major challenges include variations in wastewater composition and operational efficiency. To solve the issues, the best approach is to monitor regularly, optimize processes, and implement new, advanced technologies for treating industrial wastewater.

Overall, the effluent treatment plants prevent the discharge of untreated harmful water into the water bodies and the environment, which ultimately protects the ecosystem and public health as regular monitoring of discharge ensures the meeting of regular standards for wastewater discharge.

(https://amritawatersolutions.com/industrial-effluent-treatment-plant-manufacturer-in-jabalpur/)

Another example is CETP in Indore, with a capacity of 4 MLD (1 MLD Industrial + 3 MLD Domestic). Wastewater from connected industries is collected through pipelines and tankers. CETP has primary, secondary, and tertiary treatment facilities.

In Indore, the treated effluent from CETP is transferred to the STP for further treatment and is partially used for horticulture, fire hydrants, road cleaning, etc.

Another CETP implication is in Bhopal, in the Govindpura industrial area, with a capacity of 0.2 MLD. Here, a continuous effluent quality monitoring system is also installed, enabling continuous tracking of wastewater discharge parameters such as pH, COD, BOD, and other

sector-specific parameters, with strict compliance with regulations. In Bhopal, 26 industries are connected with this CETP (MPPCB, 2025).

As per CPCB,

- It is difficult to operate and maintain individual effluent treatment plants for small-scale industries as they have limited scale and financial capacity.
- To address this issue for the betterment of the environment, a CETP is suggested for industrial areas, as most small-scale industries are located only in industrial areas and clusters.
- In the CETP plant, wastewater from industries is collected and processed. As a result, the common effluent treatment plant costs are reduced, it's easy to maintain, and it's easy for pollution control agencies to monitor.

This CETP step in industrial clusters for wastewater treatment is a positive, successful approach to controlling pollution.

2.3 Strategy for Collective Wastewater Management

A coordinated water/wastewater management plan is necessary to ensure that industrial development occurs sustainably within the basin. The plan would include a cluster-based treatment system; mandatory pre-treatment at the industrial level; integrated wastewater reuse; and strengthened monitoring mechanisms to ensure all dischargers comply with discharge standards. A collaborative plan will help to minimize the environmental impact of wastewater generated by industries on the River System.

As per the official memorandum from the Ministry of Environment, Forest, and Climate Change, 2026, for wastewater-handling and reuse, industries must send wastewater to the treatment plant (CETP) only through a dedicated pipeline network. After treatment, this water must be piped back to the member factories for reuse in their industries.

3 ZLD Implementation

3.1 Feasibility and Roadmap for Zero Discharge

Zero Liquid Discharge (ZLD), a cutting-edge method for managing industrial wastewater that reduces or eliminates discharges into the environment, requires industries to recover as much water as possible and use it in their operations. In this manner, the suitability of implementing ZLD across different industrial sectors in the Narmada River Basin depends on the sector type, the characteristics of the wastewater generated, the scale of operations at each facility, and the availability of adequate water treatment technology. Due to stricter environmental protection regulations and the higher risk of environmental contamination, industries that produce chemicals, manufacture drugs/pharmaceuticals, dye textiles, or whose production methods result in highly contaminated effluents are more likely to be able to implement ZLD technologies.

Data collected on the districts where industrial wastewater is produced and treated indicate that many are adopting ZLD techniques. The districts reporting at least some degree of implementation of ZLD include Bhopal, Dhar, Sehore, Damoh, and Narmadapuram. These districts also report that companies and industries operate using ZLD systems. However, in Jhabua district, all H-Acid manufacturing companies reportedly utilize MEE and drying facilities to treat their effluent and thus prevent liquid waste from entering receiving water bodies. Therefore, the fact that many districts have implemented ZLD-type wastewater treatment systems provides evidence of the rapid adoption of these systems throughout the Narmada River Basin.

In addition, during informal meetings with representatives of the Madhya Pradesh Pollution Control Board (MPPCB) who were participating in field visits and public consultations, it was stated that the state government is working toward requiring ZLD for industries that produce high levels of pollutants. MPPCB representatives noted that high-strength wastewater producers are being urged to adopt various wastewater reuse/recycling systems, with increased focus on reducing liquid effluent discharged into receiving waters. While implementation rates can vary significantly across districts and industrial sectors, the trend in Madhya Pradesh is to increase the adoption of ZLD and water reuse systems.

Therefore, a staged plan outlining the steps required for ZLD implementation would be beneficial. The first step would be to identify and encourage high-polluting industries/large facilities to implement ZLD immediately. Medium- and small-scale industries could then be

incorporated through cluster-based approaches, such as CETPs, along with advanced treatment systems.

3.2 Suggestive Sector-specific ZLD guidelines

The use of ZLD varies widely across industrial sectors, so it would be beneficial to develop specific guidelines for each industry to ensure successful application. Textile, pharmaceutical, chemical, and dye manufacturing industries will have higher strength and effluent generation; hence, they can effectively utilize ZLD systems, as these systems meet very tight discharge requirements.

ZLD implementation for textile, pharmaceutical, chemical, and dye manufacturing industries should consist of the following:

- Pre-treatment to remove suspended solids and other contaminants
- Biological treatment to reduce organic load
- Additional advanced treatment processes to handle hazardous/emerging contaminants/non-biodegradable pollutants.
- Properly planned management of any type of concentrate.

On the other hand, the low-polluting nature of industries involved in agro-processing and small-scale manufacturing units could pursue partial reuse schemes or integrate with a CETP system rather than complete ZLD systems. The development of differentiation among industries will enable cost-effectiveness while ensuring that all industries remain environmentally compliant.

Zero Liquid Discharge System means that the industrial effluent after treatment is reused within the industry only, and any waste or solute is converted into a solid form using methods such as concentration/evaporation/drying and can be disposed of as per guidelines. The most important factors for ZLD implementation in any industry are water consumption versus water reused or recycled, and solid recovery from effluents. There is no specific technology recommended by CPCB for achieving ZLD, as effluent recycling varies by industry (CPCB).

Currently, sector-wise, distillery industries have adopted the zero liquid discharge system, and to achieve that, concentration/evaporation and drying/incineration/bio-composting systems are used. Many other industries, such as the paper and pulp industry, which are based on wood, have not adopted ZLD due to high investment costs. Other sectors, such as the Pharmaceutical,

tannery, and textile industries, adopted ZLD systems in accordance with SPCB guidelines (National Green Tribunal, application no. 682/2019).

3.3 Gaps in the implementation of the CETP and ZLD systems.

On the basis of the available data for CETPs in the Basin, it is concluded that due to the expansion of the industries of various categories in the basin there is strong requirement of the CETP in the industrial clusters and area, so that there is proper treatment of the industrial discharge and it should also compile with the Zero liquid discharge arrangement, as there is gap in the ZLD implementation. As per data, only a few industries in districts like Dhar, Sehore, Bhopal, Jhabua, Narmadapuram, and Bhopal show implementation of Zero liquid discharge, but many districts, such as Balaghat, Betul, Katni, Mandla, Narsinghpur, Khandwa, and Khargone, need treatment plant facilities for protecting the ecosystem from the pollutant water and waste.

There is no zero-discharge arrangement for the disposal of treated plant effluent; also, no record of pumping, effluent sent to STP, or other activities. There is no maintenance of the pipeline that carries wastewater from industries to the CETP, and a CETP plant is required in another industrial area.

4 Hazardous Waste

4.1 Treatment and Secure landfill needs

The district-wise status of hazardous waste generation, treatment, and disposal across the Narmada River Basin is presented in *Tables 2 & 3*. As can be observed from this table, many districts have reported significant amounts of hazardous waste generated, especially those industrialized areas such as; Raisen (5711.8 MT/annum), Indore (5724.57 MT/annum), Dewas (6971.18 MT/annum), Khandwa (22,969.45 MT/annum) and Sehore (9335.93 MT/annum) have reported substantial amounts of hazardous wastes requiring treatment and disposal. Districts along the upper basin, i.e., Raisen, Sagar, and Jabalpur, have reported generating some of the largest quantities of hazardous waste; however, there is very little to no available infrastructure for these districts to rely upon other than Treatment, Storage, and Disposal Facility (TSDFs) established in Pithampur (Dhar District). Similarly, in the middle basin, districts of Dewas, Dhar, Indore, and Khandwa reported

producing significant quantities of hazardous waste. However, they were all reliant on a single TSDF facility in Pithampur. This centralization of TSDFs in one area can create logistical issues with the transportation of hazardous waste and may limit available capacity for processing it.

Further analysis of the data indicates that although significant portions of hazardous waste are identified as recyclable or usable, a considerable amount still needs to be disposed of securely in landfills or incinerators. Specifically, districts such as Dewas and Khandwa reported significantly large volumes of landfillable hazardous waste, indicating an urgent need to develop additional secure landfill capacity. Additionally, the development of captive secured landfill facilities is severely underdeveloped. Only the district Jabalpur reports having developed secure landfill facilities.

Table 3:District-wise Hazardous Waste Generation, Treatment, Disposal Categories, and TSDF Availability across the Narmada Basin

*Source: District Environmental Plans 2025 for districts (*Data presented here is for the whole district)*

Basin	District	No. of Industries generating HW	Quantity of HW in the district MT/ANNUM	No. of captive/common TSDF	Location of TSDF	Contaminated sites or probable contaminated sites
Upper	ANUPPUR	36	604.27	1	TSDF site in the state (Pithampur, Dhar)	None
	BALAGHAT	28	37.47	1	TSDF site in the state (Pithampur, Dhar)	Nil
	BETUL	56	32.33	1	TSDF at Dhar	0
	CHHINDWARA	115	2862.74	1	TSDF in Pithampur, Dist. Dhar	0
	DAMOH	11	42.64	1	TSDF available at	Nil

					Pithampur, District Dhar	
	DINDORI	2	0	1	TSDF MP Wastes Management, Pithampur, District Dhar, M.P.	Nil
	NARMADAPURAM	90	66.89	1	TSDF located at Pithampur, Dhar, Madhya Pradesh	Nil
	JABALPUR	192	2566.19	2 Captive Secured Landfill Facilities; No Common TSDF	RPG Transmission Ltd. and Telecom Ltd.	None
	KATNI	71	101.8	1	TSDF site located at Pithampur, Dhar, Madhya Pradesh	Nil
	MANDLA	31	10.967	None	None	
	NARASINGHPUR	28	262.62	1	Pithampur, Dhar	Not available
	RAISEN	172	5711.8	1	TSDF in Pithampur, District Dhar	Nil
	SAGAR	52	6322.21	1	TSDF Site located at Pithampur, Dhar District, MP	No contaminated site in Sagar district
	SEONI	24	0	1	TSDF in Pithampur, Dist. Dhar	Nil
	UMARIA	31	32.48	1	Pithampur TSDF	None

Middle	ALIRAJPUR	24	0.05	1	TSDF Site located at Pithampur, District Dhar, M.P.	Nil
	BARWANI	24	314.26	1	TSDF site in the state (Pithampur, Dhar)	
	BHOPAL	556	2387	None in the district	None	1 (Union Carbide)
	BURHANPUR	47	4503.9	1	TSDF site located at Pithampur District, Dhar	None
	DEWAS	135	6597.18	1	CTSDF MP Wastes Management, Pithampur, Dist. Dhar (M.P.)	0
	DHAR	467	29210.15	1	Dhar	0
	HARDA	27	136.404	Nil	Nil	Nil
	INDORE	940	5724.57	-	-	
	JHABUA	3	1184.53	1	Pithampur Industrial Waste Management Pvt. Ltd.	Nil
	KHANDWA (E.N.)	38	62.93.45	1	TSDF Site located at Pithampur, Dhar, Madhya Pradesh	Nil
	KHARGONE (W.N.)					
	SEHORE	48	9335.93	Nil	Nil	Nil

Table 4: District-wise Hazardous Waste Generation, bifurcation.

*Source: District Environmental Plans 2025 for districts (*Data presented here is for the whole district)*

Basin	District	Quantity of incinerable HW MT/ANNUM	Quantity of land fillable HW MT/ANNUM	Quantity of Recyclable HW MT/ANNUM	Quantity of utilizable HW MT/ANNUM
Upper	ANUPPUR	3.31	0	35.89	565.07
	BALAGHAT	0	0	37.47	37.47
	BETUL	0	9.78	21.01	1.54
	CHHINDWARA	0.04	1642.13	154	0
	DAMOH	0	0	42.64	0
	DINDORI	0	0	0	0
	NARMADAPURAM	1.22	2.4	43.3	19.98
	JABALPUR	11.69	88.13	2466.38	0
	KATNI	0.1	0.19	101.51	0
	MANDLA	0	10.107	0.86	Nil
	NARASINGHPUR	0	252.44	10.18	NA
	RAISEN	173.62	2396.28	119.05	NA
	SAGAR	5.51	206.51	304.08	5805.1
	SEONI	0	0	8.25	8.25
	UMARIA	9	6.25	17.22	NA
Middle	ALIRAJPUR	0	0	0.05	0
	BARWANI	125	0	35.92	152.86
	BHOPAL	133	73	2044	135
	BURHANPUR	0	25.79	0.11	4478
	DEWAS	190.74	1475.08	4931.36	0
	DHAR	1142.17	6623.47	21144.57	0
	HARDA	135.327	Nil	0.9421	0.135
	INDORE	424.077	-	-	-
	JHABUA	1.24	121.83	960.55	100.91
	KHANDWA (E.N.)	6.46	25.80	30.67	0

	KHARGONE (W.N.)				
	SEHORE	22.69	8958.08	350.03	5.14

Table 5:Hazardous Waste Generation in the Lower Narmada Basin(*Source: Gujrat Pollution Control Board, District Environmental Plans 2022&2025).*

District	Hazardous Waste Generation (MT/year)
Bharuch	12,53,485
Narmada	312
Chhota Udepur	11.095

Table 5 provides district-wise hazardous waste generation within the Lower Narmada Basin, expressed in metric tonnes per year (MT/year). The data indicate variation in hazardous waste production across the Bharuch, Narmada, and Chhota Udepur districts, reflecting differences in industrial activities and waste-generation patterns.

The existing system in the Upper and Middle Narmada Basin is heavily dependent on the Pithampur-Dhar district TSDF, with inadequate regional distribution. This highlights the requirement for:

- Expansion of secured landfill facilities in high waste-generating districts
- Development of additional TSDF sites to reduce transportation burden
- Strengthening of waste segregation and categorization at source
- Promotion of recycling and resource recovery for recyclable fractions

4.2 Strategy for Toxic Liquid Waste

Toxic liquid waste generated by industrial processes poses a critical risk if not properly managed, as it can be directly discharged into water bodies. This hazardous waste stream may harm people and the environment. Currently, most districts manage toxic liquids using a combination of in-house treatment and/or partial recycling, along with TSDF (Treatment, Storage, and Disposal Facility) disposal. Many districts also lack advanced treatment systems to provide additional protection. Thus, to provide adequate protection against improper handling of toxic liquid waste, there should be an organized plan/strategy developed for managing toxic liquid waste, which would include:

• A mandatory pre-treatment at the industry level prior to disposal or transport of the waste
• The integration of Zero Liquid Discharge (ZLD) technologies for polluting industries
• The development of a common facility for treating liquid hazardous wastes
• The strengthening of tracking/manifesting requirements for the transportation of hazardous waste.
• The promotion of co-processing and proper disposal in permitted facilities

5 Plan for continuous monitoring

5.1 Overview of Real-Time Monitoring Systems

Deployment of real-time environmental monitoring systems within all sectors of industry in the Narmada River basin is a direct result of the increased focus on compliance, transparency, and pollution prevention. An extensive array of technologies is included in the monitoring system, including Continuous Effluent Quality Monitoring Systems (CEQMS), Continuous Emission Monitoring Systems (CEMS), Continuous Ambient Air Quality Monitoring Stations (CAAQMS), Reverse Osmosis (R.O.) systems, flow meters, and industrial pollution control (IPC) equipment.

5.1.1 Real-time monitoring of Agro & Food Industries

There is a marked trend toward greater environmental regulatory compliance through real-time monitoring of agro and food processing facilities throughout the Narmada Watershed. Most of the larger sugar and distilleries within the watershed (as shown in *Table 4*) have implemented critical components necessary to monitor their emissions and effluent quality through CEMS and CEQMS, respectively. Most of the sugar mills located in the upper part of the Narmada Watershed (e.g., the districts of Jabalpur and Narsinghpur) also employ CEQMS and flow meters.

Table 6: Real-time Monitoring of Agro & Food Industries in the Upper Narmada Basin

Industry Name	**District**	**Sector**	**R.O.**	**CAAQMS**	**CEMS**	**CEQMS**	**Flow Meter**	**IPC**	**Total**
Hanumant Sugars Pvt. Limited	Betul	Sugar	Chhindwara	0	0	1	1	1	3

Shrijee Sugar and Power (P) Limited	Betul	Sugar	Chhindwara	0	0	1	1	1	3
Ramdev Sugar (P) Limited	Hoshangabad	Sugar	Mandideep	0	0	1	1	1	3
Sagar Stone Industries	Jabalpur	Agro Based	Jabalpur	0	0	0	0	4	4
Shri Balaji Biosolutions Fuels LLP	Jabalpur	Distillery	Jabalpur	0	1	1	1	2	5
Reva Kripa Sugar (P) Limited	Jabalpur	Sugar	Jabalpur	0	0	1	1	0	2
Orient Ethanol Industries Pvt Ltd	Mandla	Distillery	Jabalpur	0	1	1	1	0	3
Aakriti Sugar Mill Pvt. Ltd.	Narsinghpur	Sugar	Jabalpur	0	0	1	1	1	3
Kareli Sugar Mills Pvt. Limited	Narsinghpur	Sugar	Jabalpur	0	0	1	2	1	4
Mahakaushal Sugar & Power	Narsinghpur	Sugar	Jabalpur	0	0	1	1	0	2
Narmada Sugar Mills	Narsinghpur	Sugar	Jabalpur	0	2	1	1	1	5

Raj Rajeshwari Sugar Pvt. Limited	Narsinghpur	Sugar	Jabalpur	0	1	1	1	0	3
Shakti Sugar Mill (P) Limited	Narsinghpur	Sugar	Jabalpur	0	0	1	1	1	3
Vanshika Sugar & Power Industries Ltd.	Narsinghpur	Sugar	Jabalpur	0	0	1	1	1	3

Data retrieved from the MPPCB website (https://erc.mp.gov.in/IMS/CRUD/R_RTM**)*

The extent of monitoring is greater for all middle-basin industrial facilities compared to those in the upper basin. For example, both Maa Rewa Sugar and Security Paper Mill exhibit greater than average numbers of monitoring system components. Similarly, distilleries in Khargone are equipped with two CEMS (continuous emission monitoring systems) each, as shown in *Table 5*. Therefore, there appears to be a significant difference in compliance levels among these facilities relative to their upper-basin counterparts.

Table 7:Real-time Monitoring of Agro & Food Industries in the Middle Narmada Basin

Industry Name	**District**	**Sector**	**R.O.**	**CAAQMS**	**CEMS**	**CEQMS**	**Flow Meter**	**IPC**	**Total**
Maa Rewa Sugar Pvt. Limited	Dhar	Sugar	Dhar	0	1	1	1	1	4

Security Paper Mill	Hoshangabad	Agro Based	Mandideep	0	5	1	1	1	8
Agrawal Distilleries Pvt. Limited	Khargone	Distillery	Indore	0	1	0	3	1	5
Associated Alcohal & Breweries Ltd.	Khargone	Distillery	Indore	0	2	0	1	2	5
Associated Alcohols And Breweries Ltd. (Ethanol Plant)	Khargone	Distillery	Indore	0	1	0	1	1	3
Agrawal Distilleries Pvt Ltd. Unit 2 (Grain Base Distillery with Power Plant)	Khargone	Distillery	Indore	0	1	0	1	0	2

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

5.1.2 Real-time monitoring of the Manufacturing sector

Monitoring real-time emissions in the manufacturing sector clearly illustrates this distinction. As shown in *Table 6*, industries in the upper basin have limited monitoring systems, with only basic installations such as flow meters, IPC devices, and occasional CEMS units.

This indicates minimal but functional compliance with environmental monitoring requirements.

Table 8: Real-time Monitoring of the Manufacturing Sector in the Upper Narmada Basin

Industry Name	**District**	**Sector**	**R.O.**	**CAAQMS**	**CEMS**	**CEQMS**	**Flow Meter**	**IPC**	**Total**
Narmada Gelatines Limited	Jabalpur	Manufacturing	Jabalpur	0	0	0	0	2	2
Perfect Sonewar e Pipes Ltd.	Mandla	Manufacturing	Jabalpur	0	1	0	0	0	1

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

The middle basin demonstrates significantly stronger monitoring coverage. Textile and cement industries, particularly in districts such as Sehore and Dhar, are equipped with multiple monitoring systems, including CAAQMS, CEQMS, and flow meters. Large units such as Ultratech Cement and Sagar Cements have the highest number of installed devices, reflecting stricter regulatory compliance and higher pollution control requirements shown in *Table 7*.

Table 9: Real-time Monitoring of the Manufacturing Sector in the Middle Narmada Basin

Industry Name	**District**	**Sector**	**R.O.**	**CAAQMS**	**CEMS**	**CEQMS**	**Flow Meter**	**IPC**	**Total**
Trident Corporation Limited	Sehore	Textile	Bhopal	0	0	0	2	4	6
Trident Limited (Bed Linen Division)	Sehore	Textile	Bhopal	1	1	0	0	0	2

Sagar Cements (M) Private Limited - (Old Name- Satguru Cement Pvt. Ltd.)	Dhar	Cement	Dhar	2	5	0	0	0	7
Ultratech Cement Ltd_Dhar	Dhar	Cement	Dhar	4	9	0	0	1	14
Pioneer Industries (Old Name - Satguru Industries)	Dhar	Cement	Bhopal	0	1	0	0	0	1

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

5.1.3 Real-time monitoring of Thermal Power Plants

Thermal power units in the Upper Basin maintain strong real-time monitoring coverage, as summarized in *Table 8*. Super Thermal Power Station at Narsinghpur has the highest level of monitoring, with 10 sensors across CAAQMS, CEMS, and CEQMS. Satpura Thermal Power Station in Betul also has the monitoring level with 7 sensors, including a flow meter.

Table 10: Real-time Monitoring of Thermal Power Plants in the Upper Narmada Basin

Industry Name	**District**	**Sector**	**R.O.**	**CAAQMS**	**CEMS**	**CEQMS**	**Flow Meter**	**IPC**	**Total**

Satpura Thermal Power Station (MPPGCL- 2 X 250 MW), Betul	Betul	Thermal Power	Chhindwara	0	2	0	0	0	2
Satpura Thermal Power Station (New Ash Pond of 111 Hectare), Betul	Betul	Thermal Power	Chhindwara	2	0	1	1	3	7
BLA Power (P) Limited	Narsinghpur	Thermal Power	Jabalpur	1	1	0	0	1	3
Super Thermal Power Station	Narsinghpur	Thermal Power	Jabalpur	4	2	1	0	3	10
Jhabua Power Limited	Seoni	Thermal Power	Jabalpur	1	1	1	0	1	4

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

Monitoring thermal power-generating stations in the middle basin demonstrates that all listed stations are subject to significant real-time monitoring. As shown in Table 9, NTPC Khargone has the most devices installed to monitor its station, with a total of 13 sensors, including CAAQMS, CEMS, and CEQMS. The Singaji Thermal Power Project in Khandwa has 6 sensors. Additionally, Vardhman Fabrics (CPP) is typical for captive power stations in terms of compliance. The various systems provide continuous real-time data to regulatory agencies.

Table 11: Real-time Monitoring of Thermal Power Plants in the Middle Narmada Basin

Industry Name	District	Sector	R.O.	CAAQMS	CEMS	CEQMS	Flow Meter	IPC	Total
NTPC Limited	Khargone	Thermal Power	Indore	4	4	1	0	4	13
Shree Singaji Thermal Power Project, Stage-1, M.P. Power Generating Co.	Khandwa	Thermal Power	Indore	2	2	0	1	1	6
Shri Singaji Taap Vidyut Pariyojna Stage-II	Khandwa	Thermal Power	Indore	1	2	0	0	1	4
Vardhman Fabrics (CPP) 24 MW	Sehore	Thermal Power	Bhopal	1	1	1	1	2	6

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

5.1.4 Real-time monitoring of water & waste management

Upper Basin Waste & Water Treatment Facilities have installed a selective variety of real-time monitoring technologies, as shown in *Table 10.* At Jabalpur's 32 MLD STP, CEQMS & flow meter technologies are installed to continuously monitor the quality of discharged effluents.

Table 12: Real-time Monitoring of Water and Waste Management Facilities in the Upper Narmada Basin

Industry	District	Sector	R.O.	CAAQMS	CEMS	CEQMS	Flow Met	IP	Total

Name							er	C	
Elite Engineers	Jabalpur	CBWTF	Jabalpur	0	1	0	0	0	1
Jabalpur MSW Pvt. Limited	Jabalpur	CMSWTF	Jabalpur	0	1	0	0	0	1
Bhedaghat Sewer Project Jabalpur (0.43 MLD STP - Panchwati)	Jabalpur	STP	Jabalpur	0	0	1	0	0	1
The Commissioner, Nagar Palik Nigam, Jabalpur, 32 MLD STP	Jabalpur	STP	Jabalpur	0	0	2	2	0	4

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

Middle Basin water and waste treatment facilities all include structured monitoring systems, as shown in *Table 11*. Multiple CEQMS and CEMS technologies were utilized at Khargone STP (17.60 MLD) to ensure compliance with effluent regulations. Additionally, CEMS is used by Medisure Incinerators in Khandwa for continuous air quality monitoring during their biomedical waste disposal operations. Both systems promote environmental compliance and provide a basis for the development of basin-wide pollution control strategies.

Table 13: Real-time Monitoring of Water and Waste Management Facilities in the Middle Narmada Basin

Industry Name	District	Sector	R.O.	CAAQMS	CEMS	CEQMS	Flow Meter	IPC	Total

Nagar Palika Parishad Khargone (17.60 MLD STP)	Khargone	STP	Indore	0	0	2	2	0	4
Medisure Incinerators	Khandwa	CBWTF	Indore	0	1	0	0	0	1

*Data retrieved from the MPPCB website (**https://erc.mp.gov.in/IMS/CRUD/R_RTM**)

5.2 Gaps in Monitoring Infrastructure

Although numerous monitoring systems are used across a variety of sectors, many of these have technology that can be monitored. There are still gaps in the monitoring system's use and application, particularly regarding how consistently these systems monitor industrial operations and which technologies are deployed within them. There is also evidence that monitoring systems are generally found in larger, higher-pollution-producing industries. Smaller and lower pollution-producing industries and/or scattered (dispersed) production clusters may not have an adequate or real-time monitoring infrastructure. Also, because of differences in the numbers and types of monitoring devices deployed across districts, it appears that monitoring systems have been implemented inconsistently.

6 Policy Interventions

- Strengthening regulatory enforcement and implementing new policies to promote Zero Liquid Discharge (ZLD) are key elements of an industrial wastewater management plan that will strengthen the existing Central Effluent Treatment Plant (CETP) and Cluster-based infrastructure in the Narmada Basin and improve hazardous waste management systems.
- Financial Incentives, Institutional Mechanisms, and Public-Private Partnerships will support the adoption of Real Time Monitoring Technologies.
- A coordinated effort at the level of the entire basin, through central monitoring and collaboration among multiple agencies at the district level, will be necessary to implement this strategy uniformly throughout all districts of the basin.

- Collectively, these strategies should provide a balance between economic growth/industrialization and environmental conservation within the Narmada Basin.

Details of industrial wastewater governance are shown in *Figure 2.*

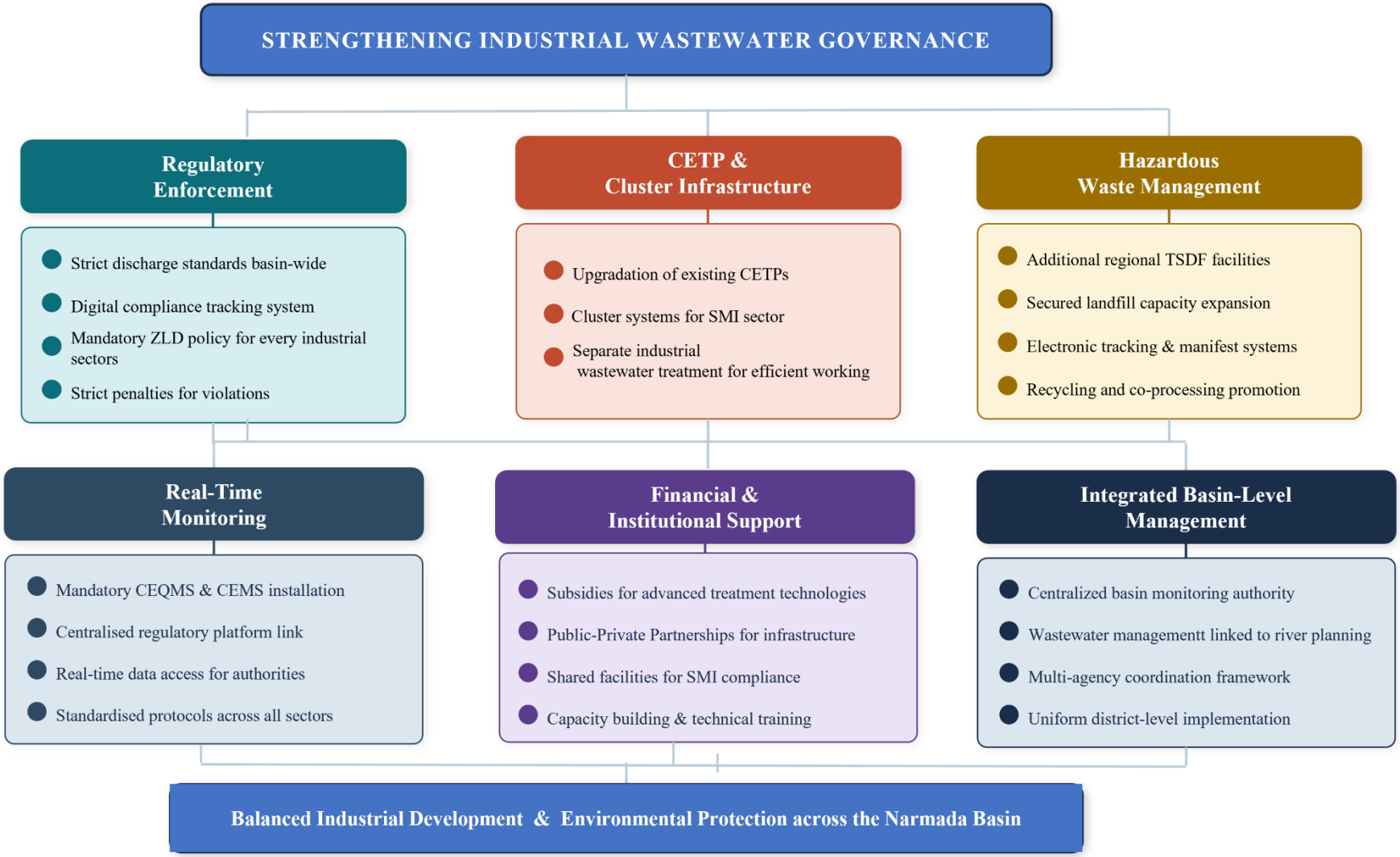


Figure 2: Industrial Wastewater Governance Framework for the Narmada Basin

7 Role of Stakeholders

The implementation of the industrial wastewater management strategy in the Narmada Basin will depend upon how all relevant stakeholders work together to provide their respective inputs into the areas of governance, infrastructure, and regulatory compliance.

- Regulatory agencies will be able to set standards for water quality, monitor and enforce the compliance of these standards by industry, and enforce all applicable environmental regulations.
- Industries will need to properly treat their effluent, adopt Zero Liquid Discharge (ZLD) technologies, and safely handle any hazardous waste generated. Associations of industries and cluster organizations can help build common infrastructure, for example, Central Effluent Treatment Plants (CETPs), and assist small businesses in meeting compliance requirements.
- Urban local government entities and district administration entities will have a role in providing land for industry and in creating plans for needed infrastructure that integrate with area-wide development initiatives.
- TSDF (treatment storage/disposal facility) operators manage or dispose of hazardous waste safely. Financial Institutions and Private-Sector Entities support the development of Treatment Infrastructure through funding and/or partnering with treatment facilities.
- Research Institutions and Technical Agencies provide additional support to the System by developing new technologies, improving capacities for treating hazardous wastes, and evaluating the performance of treatment processes.

A coordinated Multi-Stakeholder Approach, as shown in *Figure 3*, will be necessary to implement these systems effectively, thereby ensuring compliance, enhancing infrastructure, and promoting the sustainable use of resources within the Basin.

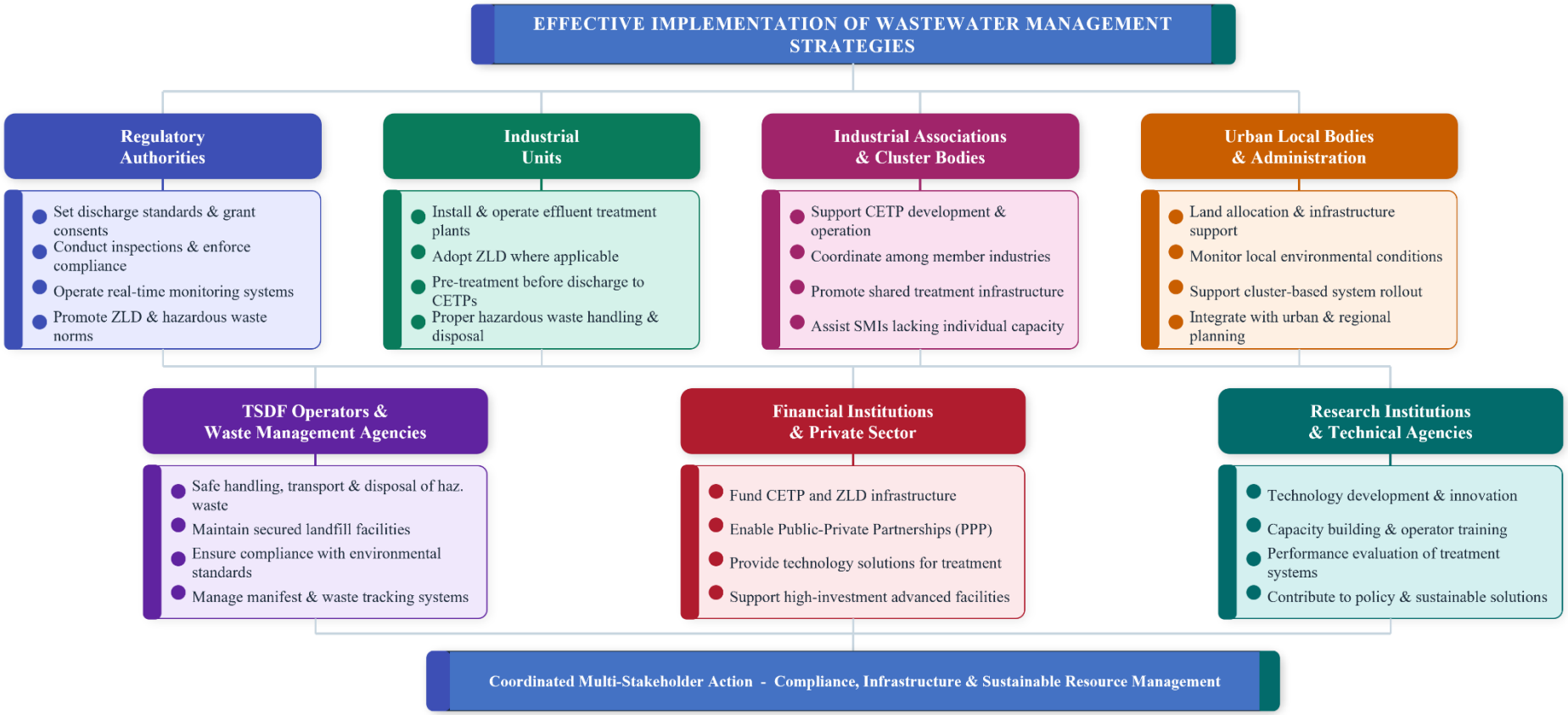


Figure 3: Stakeholder Framework for Effective Implementation of Industrial Wastewater Management in the Narmada Basin

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